1. Background/Context of the Business Scenario

The SCQA problem-solving framework will be used to break down the NHS business scenario.

Situation: The NHS must decide whether to expand infrastructure and capacity or optimise use of existing resources. This is a decision that needs to be data-informed and not opinion-driven.

Complication: Some stakeholders disagree on whether current capacity is inadequate or simply underused, and costly missed GP appointments remain poorly understood.

Question: Two core business questions frame this analysis: has there been adequate staff and capacity in the networks, and what was the actual utilisation of resources?

Answer: This analysis investigates GP appointment data to answer both questions using evidence rather than opinion, with a secondary exploration of whether Twitter/X data adds further decision-making value.

2. Analytical Approach

Python was used as the analytical tool for performing data wrangling and manipulation because of its powerful libraries like pandas, NumPy, and matplotlib. These allow efficient cleaning, analysis and visualisation of the large NHS dataset with accuracy.

2.1 Dataset Schema

Three datasets were imported using pandas, which include actual_duration.csv, appointments_regional.csv, and national_categories.csv. They were imported into separate DataFrames (ad, ar, nc) rather than combined immediately, since their structures differ. Initial exploration used .head(), .info(), .describe(), and .isnull().sum() to establish shape, data types, and completeness before any analysis began. This step was treated as a validation gate. Using .isnull() across all three datasets revealed no true NaN values, but cross-checking with .value_counts() showed a more significant issue: disguised missingness, where categorical fields contained literal "Unknown," "Unmapped," or "Inconsistent Mapping" values rather than nulls. This shaped the entire analytical approach, since it meant standard null-handling methods were insufficient; each affected field required explicit value_counts() auditing to quantify data quality before any figure derived from it could be trusted.

[108]:
```
# Import the national_categories data set

nc = pd.read_csv('national_categories.csv')
nc.head()
```

[108]:

	appointment_date	icb_ons_code	sub_icb_location_name	service_setting	context_type	national_category	count_of_appointments	appointment_month
0	02/08/2021	E54000050	NHS North East and North Cumbria ICB - 00L	Primary Care Network	Care Related Encounter	Patient contact during Care Home Round	3	2021-08
1	02/08/2021	E54000050	NHS North East and North Cumbria ICB - 00L	Other	Care Related Encounter	Planned Clinics	7	2021-08
2	02/08/2021	E54000050	NHS North East and North Cumbria ICB - 00L	General Practice	Care Related Encounter	Home Visit	79	2021-08
3	02/08/2021	E54000050	NHS North East and North Cumbria ICB - 00L	General Practice	Care Related Encounter	General Consultation Acute	725	2021-08
4	02/08/2021	E54000050	NHS North East and North Cumbria ICB - 00L	General Practice	Care Related Encounter	Structured Medication Review	2	2021-08

[13]:
```
# check data types
nc.dtypes
```

[13]:
```
appointment_date         object
icb_ons_code             object
sub_icb_location_name    object
service_setting          object
context_type             object
national_category        object
count_of_appointments     int64
appointment_month        object
dtype: object
```

[14]:
```
# metadata of the data set
nc.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 817394 entries, 0 to 817393
Data columns (total 8 columns):
 #   Column                Non-Null Count   Dtype 
---  ------                --------------   ----- 
 0   appointment_date      817394 non-null  object
 1   icb_ons_code          817394 non-null  object
 2   sub_icb_location_name  817394 non-null  object
 3   service_setting       817394 non-null  object
 4   context_type          817394 non-null  object
 5   national_category     817394 non-null  object
 6   count_of_appointments  817394 non-null  int64 
 7   appointment_month     817394 non-null  object
dtypes: int64(1), object(7)
memory usage: 49.9+ MB
```

[15]:
```
# Check for missing values

nc_na = nc[nc.isna().any(axis = 1)]
nc_na.shape
```

[15]:
```
(0, 8)
```

The output shows that there are no rows with NAN in 8 columns.

[32]:
```
# Check for 'Unknown' style values hiding as valid categories

print("actual_duration value counts:")
print(ad['actual_duration'].value_counts())

print("\nappointment_status value counts:")
print(ar['appointment_status'].value_counts())

print("\nhcp_type value counts:")
print(ar['hcp_type'].value_counts())

print("\nappointment_mode value counts:")
print(ar['appointment_mode'].value_counts())
```

```
actual_duration value counts:
actual_duration
Unknown / Data Quality    20161
1-5 Minutes               19909
6-10 Minutes              19902
11-15 Minutes             19738
16-20 Minutes             19534
21-30 Minutes             19452
31-60 Minutes             19097
Name: count, dtype: int64

appointment_status value counts:
appointment_status
Attended    232137
Unknown     201324
DNA         163360
Name: count, dtype: int64

hcp_type value counts:
hcp_type
Other Practice staff    241557
GP                      226036
Unknown                 129228
Name: count, dtype: int64

appointment_mode value counts:
appointment_mode
Face-to-Face    180410
Telephone       166483
Home Visit      111194
Unknown          79147
Video/Online     59587
Name: count, dtype: int64
```

2.2 Data to Insight Translation

Business questions were broken into analytical tasks. "Is capacity adequate?" was translated into comparing appointment volume against the stated NHS benchmark (1,200,000/day); "what is actual utilisation?" was broken into service setting, context type, and national category breakdowns; "why are appointments missed?" required isolating and volume-weighting appointment_status data frame, since row-count and appointment-volume proportions showed substantial difference.

2.3 Data Wrangling

The dates in each dataset required careful handling as each dataset used a different format and start date. actual_duration and national_categories were converted using pd.to.datetime format strings to avoid day/month ambiguity. Date range profiling showed the three datasets only fully overlapped from December 2021 to June 2022, so cross-dataset comparisons were scoped to this window; appointments_regional's longer history (from January 2020) was used only where seasonality across multiple years was being assessed.

```
# Identify the earliest and latest appointment dates

# Date range in actual_duration.csv
print(f"actual_duration date range: {ad['appointment_date'].min()} to {ad['appointment_date'].max()}")

# Date range in national_categories.csv
print(f"national_categories date range: {nc['appointment_date'].min()} to {nc['appointment_date'].max()}")

# Month range in appointments_regional.csv
print(f"appointments_regional month range: {ar['appointment_month'].min()} to {ar['appointment_month'].max()}")
```

```
actual_duration date range: 2021-12-01 00:00:00 to 2022-06-30 00:00:00
national_categories date range: 2021-08-01 00:00:00 to 2022-06-30 00:00:00
appointments_regional month range: 2020-01-01 00:00:00 to 2022-06-01 00:00:00
```

Finding: The three datasets cover different, overlapping date ranges:

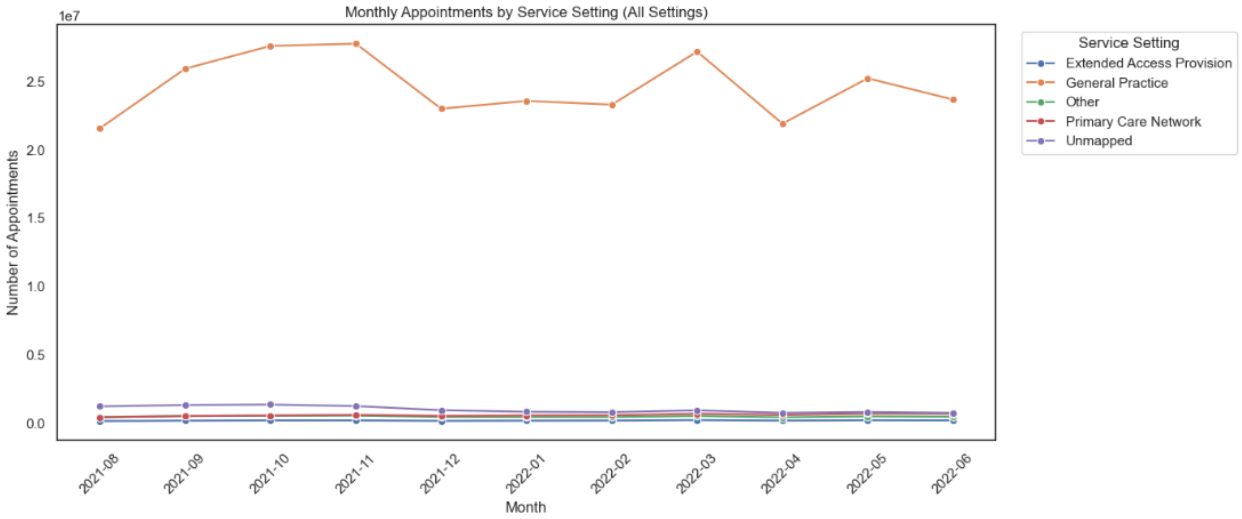
- `actual_duration` : 2021-12-01 to 2022-06-30 (7 months)
- `national_categories` : 2021-08-01 to 2022-06-30 (11 months)
- `appointments_regional` : 2020-01-01 to 2022-06-01 (29 months, month-level only)

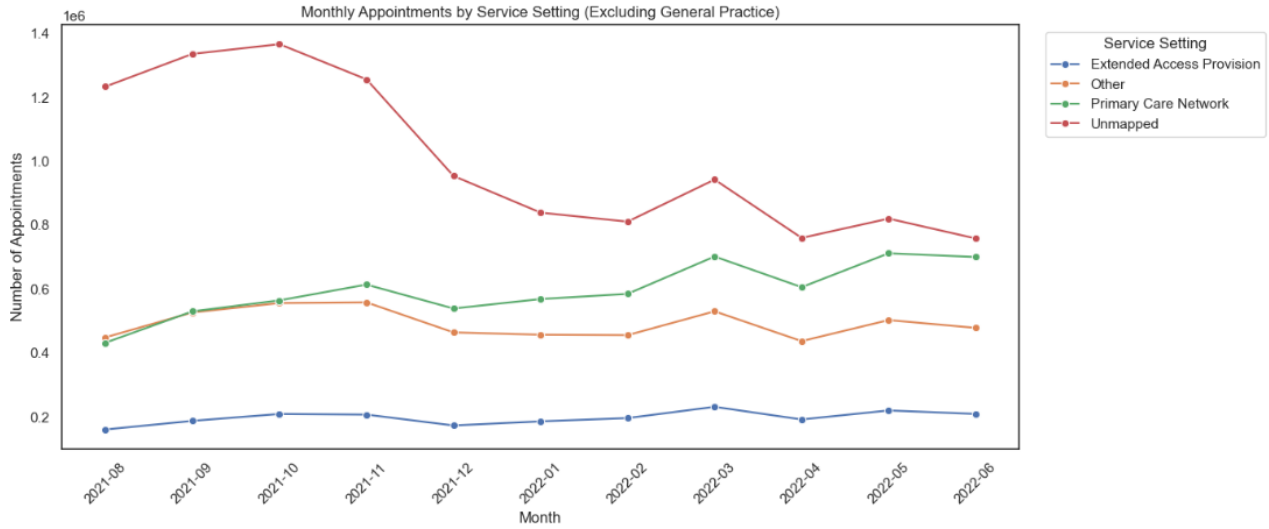
Instead of merging all datasets into a single table, each was kept separate and linked through shared dimensions such as region and month. This avoided information-reducing aggregation, since appointments_regional reports at ICB level while the other datasets report at sub-ICB level. Because each row already represents an aggregated count, the analysis relied on grouping and summarising operations such as groupby(), .sum(), and .reset_index().

3. Visualisation and Insights

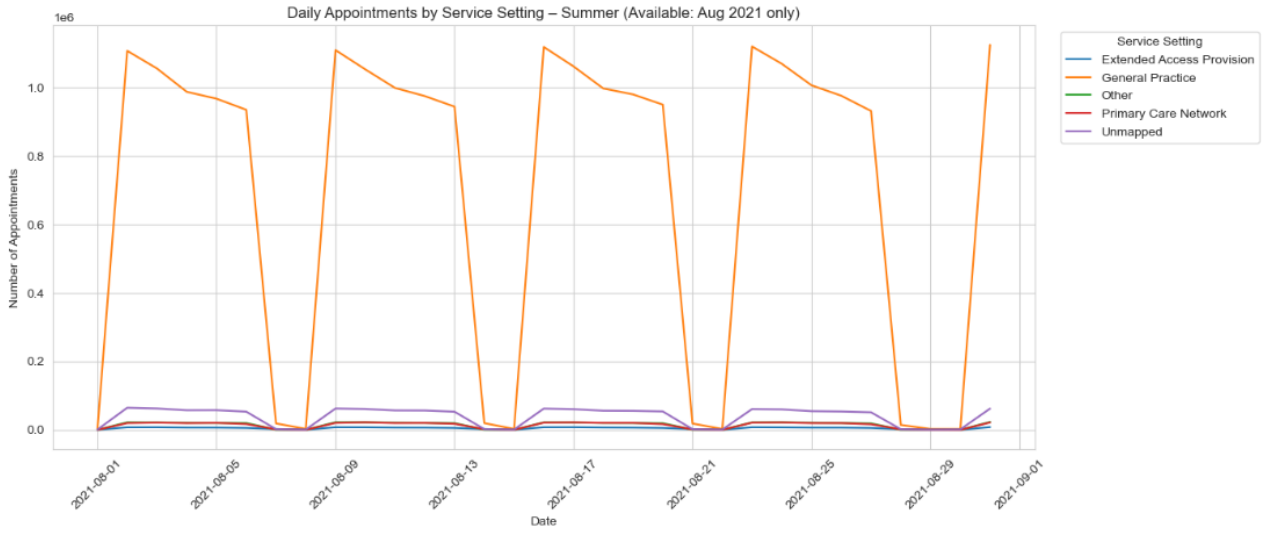
Visualisations were used in two distinct modes: exploratory, to interrogate the data and show patterns during analysis, and explanatory, to communicate a specific, pre-identified finding clearly to stakeholders. This distinction shaped every design decision in the notebook.

Seaborn lineplot() was the primary exploratory tool for monthly trends across service setting, context type, national category, HCP type, appointment status, appointment mode, and booking lead time. Line charts were chosen because the business questions are fundamentally about change over time. However, an early limitation was noted as General Practice's scale (tens of millions monthly) shadowed every other category, compressing smaller service settings into an unreadable flat line near zero. This was addressed by producing a second, filtered chart excluding General Practice, as the first version was misleading about the smaller categories' behaviour.





Another significant visualisation decision was moving from monthly-aggregated line charts to daily-level seasonal charts (Activity 4, Objective 2 in notebook). Monthly charts alone would have supported a conclusion that utilisation stayed comfortably under the NHS's 1,200,000/day benchmark. Daily-level plotting revealed a strong, repeating weekly sawtooth pattern which showed that appointments were concentrated on weekdays but were near-zero on weekends. This was invisible at monthly resolution and shows a clear case where in-depth exploration not only changed the analytical presentation but also its conclusion.



Boxplots were introduced for Activity 6, Question 6, as an explanatory complement to the line charts. Rather than showing *when* volume changed, they show *how consistent* each service setting's monthly volume was, revealing that "Unmapped", which is not a real category, was the largest non-GP setting by volume, a finding a line chart alone would not have isolated as cleanly.

For the business presentation specifically, charts were further refined: legends moved outside plot areas, titles made specific rather than generic, and the top-6-category filter applied to the crowded 18-category national category chart.



4. Patterns and Recommendations

Three patterns emerged with high confidence, each cross-validated across multiple independent breakdowns.

First, a consistent weekly cycle drives appointment volume, with near-zero weekend activity, meaning capacity should be assessed against weekday load, not monthly averages.

Second, a seasonal peak (October - November, recurring in March) appears across every dimension tested, likely reflecting genuine pre-winter demand; the NHS should plan flexible seasonal staffing around these windows.

Third, disguised missingness (Unknown/Unmapped values) affects up to a third of key fields, meaning fields such as “Did Not Attend” rate (4 - 5% by volume, not 27% by row count) and service-setting mix remain partially obscured.

Directly answering the NHS’s core questions, capacity didn’t seem adequate during peak periods. Monthly-average utilisation peaked at around 1.01M appointments/day against the 1.2M benchmark, comfortably under. But adjusting for the weekly cycle, weekday-equivalent demand during October - November and March has the possibility of exceeding 1.2M/day, since the same volume compressed into 22 weekdays rather than spread across 30 calendar days pushes the true figure roughly 15 - 35% higher.

On utilisation, “General Practice” heavily dominates the majority of appointments, with "Unmapped" obscuring the true second-largest setting.

Another recommendation is for the NHS to target additional weekday capacity specifically for peak months (October – November and March), and to also prioritise fixing data capture for fields inputted as “Unmapped”, “Unknown”, and “Inconsistent Mapping”.

Appendix

Appendix A - Data Preparation and Validation Detail

A.1 Dataset schema and scale

Dataset	Rows	Columns	Granularity
actual_duration.csv	137,793	8	Sub-ICB location, daily
appointments_regional.csv	596,821	7	ICB-level only, monthly
national_categories.csv	817,394	8	Sub-ICB location, daily

Key columns by dataset:

actual_duration - sub-ICB location, appointment date, actual duration band, count of appointments.

appointments_regional - ICB code, appointment month, appointment status, HCP type, appointment mode, booking-to-appointment lead time, count of appointments.

national_categories - sub-ICB location, appointment date, service setting, context type, national category, count of appointments.

A.2 Initial completeness check

.isnull().sum() returned zero missing values across all columns in all three datasets. This result was not taken at face value (see A.3).

A.3 Disguised-missingness audit (row-level proportions)

.value_counts() auditing on categorical fields found the following non-substantive category values, none of which were flagged by the null check above:

- appointment_status (appointments_regional): Attended 38.9%, Unknown 33.7%, DNA 27.4%
- hcp_type (appointments_regional): Unknown 21.7%
- appointment_mode (appointments_regional): Unknown 13.3%
- actual_duration (actual_duration): "Unknown / Data Quality" 14.6% - the single largest band in the column
- service_setting / context_type (national_categories): "Unmapped" and "Inconsistent Mapping" present alongside genuine categories

A.4 Row-count vs volume-weighted proportions

Because each row is a pre-aggregated count rather than one row per appointment, row-count proportions and appointment-volume proportions diverge. Recalculating appointment_status weighted by count_of_appointments (summed, not counted) shows DNA at approximately 4 - 5% of total appointment volume, against Attended at roughly 90%+ which is substantially different from the 27.4% row-count figure. This distinction directly informed the corrected DNA finding presented to business stakeholders.

A.5 Date formats and ranges

Dataset	Format	Min date	Max date
actual_duration	%d-%b-%y (e.g. "01-Dec-21")	2021-12-01	2022-06-30
national_categories	%d/%m/%Y (e.g. "02/08/2021")	2021-08-01	2022-06-30
appointments_regional	%Y-%m (month-level)	2020-01-01	2022-06-01

All three datasets converge on the same end date (June 2022), supporting a shared extraction cutoff. The common overlapping window used for cross-dataset comparison is December 2021 to June 2022.

Appendix B - Data Exploration & Visualisation

B.1 Monthly aggregated charts (exploratory)

Chart	Dimension	Key figures
Service setting	service_setting	General Practice 86.5% of NWL volume (Jan - Jun 2022 subset); dominates every month
Context type	context_type	Care Related Encounter around 20–26M/month vs Inconsistent Mapping/Unmapped combined at 3 - 4M/month
National category (18 categories)	national_category	General Consultation Routine dominant, 7.8M - 10.05M/month
Top-6 national category (filtered)	national_category	Planned Clinical Procedure anomaly: spike to 3.9M in Oct 2021, dropping to 2.1M by Dec 2021

B.2 Daily-level seasonal charts (exploratory to explanatory)

All four seasonal charts (summer/autumn/winter/spring) confirmed a consistent weekly cycle. Notable deviations: December 2021 holiday low (daily low of about 0.78M vs typical weekly low); early April 2022 shallow peak (1.08 - 1.1M/day vs surrounding weeks of 1.3M/day), possibly linked to the Easter bank holiday period. Summer chart limited to August 2021 only, as national_categories has no data before 2021-08-01.

B.3 Capacity utilisation chart (explanatory)

Monthly average daily utilisation (monthly total / 30) ranged from 0.80M (Aug 2021) to 1.01M (Oct - Nov 2021), against the NHS benchmark of 1.2M/day and never breached on this metric.

B.4 Visualisation limitations noted during analysis

- Line charts with >6 hue categories (18-category national category chart) become visually unreliable for distinguishing individual trends; mitigated by filtering to top 6.
- Monthly-resolution charts cannot detect within-month (weekly) patterns; required daily-level charts to surface the weekday/weekend cycle.
- Boxplots on monthly-aggregated data (11 data points per category) have limited statistical power for outlier detection; used descriptively.

Appendix C - Supporting Detail for Patterns, Predictions, and Recommendations

C.1 Utilisation - what "General Practice dominance plus Unmapped" means in practice

General Practice accounts for most of the appointment volume in every monthly and seasonal breakdown tested. What remains unresolved is the composition of the remainder: "Unmapped" as this is the single largest non-GP service setting by volume, larger than Primary Care Network, Other, and Extended Access Provision combined. This means the NHS cannot currently state how its second-tier services (beyond core General Practice) are actually being used, not because the demand isn't there, but because the data doesn't yet say where it's going.

C.2 Obstacles encountered and how they were addressed

- **Zero nulls masked real data quality issues.** Standard .isnull() checks passed cleanly across all three datasets. This was only caught by separately auditing .value_counts() on categorical fields.
- **Row-count statistics misrepresented true scale.** DNA appeared to affect 27% of records by row count, but only 4 - 5% by appointment volume. Any report or policy built on the row-count figure would have overstated the missed-appointments issue.

C.3 Recommendations for further exploration

- Extend the analysis with a second year of data to confirm the Oct - Nov/March pattern recurs annually rather than reflecting one season.
- Link “time_between_book_and_appointment” directly to “appointment_status” to test whether longer booking waits statistically correlate with higher DNA rates during peak weekday capacity periods specifically.
- Fix data capture for fields inputted as “Unmapped/Unknown” as this would help establish the NHS's true secondary-capacity picture.